JIM DAOU
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CPSC 6420 ASSIGNMENT 2

(A) **Environment description**. Describe the environment's goal, state/observation space, action space, transition dynamics, reward structure, and termination conditions. Explain any state discretization or environment settings you use.

a. The environment goal is for the agent to reach the goal(G) from the start(S) while avoiding tiles that have holes(H)
b. Observation space is 64 in the 8x8 map
c. Action space is 4; agent can move up, down, right, and left
d. Transition dynamics of frozen lake depend on if we have defined 'is_slippery'. With it set to 'True', the default behavior causes the agent to only have a 1/3 chance of moving in its intended direction and 1/3 chance of moving in either direction perpendicular to its intended direction. With it set to 'False', the agent always moves in its intended direction.
e. Reward structure for the agent is +1 for reaching the goal and 0 for everything else, including moving to a tile with a hole
f. Termination happens when the agent moves to a tile with a hole or reaches the goal tile. Episode is truncated at 200 steps for the 8x8 environment map.
g. No further state discretization settings or environment settings were used
h. The observations stated here were gathered using 'inspect_frozen_lake.py' and official Frozen Lake documentation